# Computational Complexity for Data Science

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## Welcome

Computational complexity asks how much time, memory or randomness a problem needs, whatever algorithm solves it. Data science meets it whenever fitting a model exactly turns out to be computationally hard, which is one reason practical methods use greedy search or a convex surrogate loss.

These notes collect the computational complexity that data science courses use. They are new. The pages written so far are:

- Asymptotic notation: big-O, big-Omega and big-Theta, and what it means for a computation to cost $O(D)$
- P and NP: decision problems, polynomial running time, and the classes P and NP
- Reductions and NP-completeness: polynomial-time reductions, NP-complete problems, and why sparse regression is computationally hard
- Kolmogorov complexity: the length of the shortest program that prints a string, why most strings are incompressible and no program computes it, and how its average per symbol approaches Shannon entropy

The planned topics still to come are:

- models of computation: Turing machines and computability
- space complexity
- randomized computation
- communication and information complexity

Information theory has its own library, *Information Theory for Data Science*. Results that connect the two live in the library that holds their main statement, and the other links to them.

## Starting materials

These notes are written fresh, but they draw on the following books and Stanford course pages.

- Sanjeev Arora and Boaz Barak, *Computational Complexity: A Modern Approach*, the main starting point for complexity theory.
- CS154: Introduction to the Theory of Computation (Omer Reingold and Li-Yang Tan), which covers automata, computability and complexity, using Michael Sipser's *Introduction to the Theory of Computation*, with lecture slides and videos.
- CS154: Introduction to Automata and Complexity Theory (Spring 2009, with Rajeev Motwani among the course staff), with handouts, homework and a reading list.
- *Introduction to Automata Theory, Languages, and Computation*, Jeff Ullman's home page for the book, which is the textbook the 2009 CS154 page links. It has solutions to starred exercises, errata, slides, and past CS154 course materials.
- CS254: Computational Complexity (Li-Yang Tan), which covers space complexity, the polynomial hierarchy, randomized and non-uniform computation, and interactive proofs. It uses Arora and Barak and Avi Wigderson's *Mathematics and Computation*, and links a reading for each lecture.

## Using these notes in another site

Course sites link to these pages by URL; they do not include this repository as a git submodule. A host site that keeps a copy of this repository at its root, named `cds`, can still include fragments with paths that start with `cds/`. This site includes its own fragments the same way, through a `cds` symlink that points at the repository root.

Quarto resolves `@id` cross-references only within one rendered page, so a host site that links to a result here uses an explicit link to the page and anchor.